AUtoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 1: Data and Model Selection (Revised)

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Summary

I prepared 1,999 source-linked question–answer records from public Alabama Medicaid documents, split them by document, and fine-tuned four local language-model families. Each fine-tuned adapter was compared with its own untouched base model on the same 24 held-out validation questions. No adapter showed a reliable gain: three adapters each gave 4 fully supported answers out of 24, and no paired difference is statistically reliable. I selected Nemotron 3.5 Lightning (checkpoint 400) for the one-time final test because it gives the cleanest test of what fine-tuning adds, it is the only candidate with complete outputs and a human-reviewed sample, and it answers in about 3 seconds. Everything here is development evidence; the final test has not been run.

1. Data Characteristics

The task is to answer a public Alabama Medicaid policy or billing question and give the supporting document and page, or to decline when the documents do not support an answer. The system is decision support for an analyst. It does not make eligibility, payment, clinical or legal decisions.

Item	Prepared data
Source snapshot	138 public PDF files, 2,433 pages (Administrative Code and July 2026 Provider Manual), downloaded September 14, 2026. Each file is fingerprinted with SHA-256.
Final shape	1,999 answerable question–answer records from 136 files: 750 Administrative Code and 1,249 Provider Manual. Each record keeps its source URL, page and file fingerprint.
Target	A short answer grounded in one source page (median 18 words; 95th percentile 52) plus its locator. There is no class label. The data contains no unanswerable examples; one intentionally unanswerable record was set aside.
Engineered inputs	Question plus a fixed instruction to cite the source; training loss on the answer only; nested training subsets of 200, 509, 750 and 1,600 records, each covering all 113 training documents.
Rights and privacy	Public documents only. No patient, claims or protected health information.

Split strategy: grouped by document. Every page of a PDF goes to the same split: 1,684 train / 124 validation / 191 test records from 113 / 10 / 13 documents, with zero documents shared. Manual pages repeat wording and tables from neighboring pages, so a random split would put near-copies of test pages into training and inflate scores. Assignment uses a hash of the document identity, so it is reproducible.

Changes since the Week 4 gate. (1) I excluded the September 16–18 Muse runs because they used earlier data-generation and training methods; all current runs use the amended generation, split and training procedure. (2) I added the nested-size design and completed 16 runs between September 29 and October 1. (3) The 191 historical test records were used for diagnostics, so they are no longer an untouched test; the final test will use questions from fresh official documents outside all three splits. (4) I ruled on every cross-split duplicate answer (Section 6).

2. Candidates Considered

The declared family is deep learning: low-rank adaptation (LoRA) fine-tuning of open-weight language models that run on the project's local GPU server. All four families were trained on identical nested subsets and checked against the same 124 validation records; within each run the checkpoint with the lowest validation loss was kept.

Candidate	Why it was plausible for this data and this decision
Nemotron 3.5 Lightning 30B-A3B (BF16 weights, LoRA rank 16)	Mixture-of-experts model with about 3 billion active parameters per token, so answers stay fast on one GPU. BF16 weights fit on the project's 96 GB GPU without quantization, which avoids a lossy conversion step.
Gemma 4 26B-A4B (BF16, LoRA rank 4)	A second mixture-of-experts design (about 4 billion active parameters) with a different tokenizer and training lineage. Its small rank-4 adapter tests whether light adaptation is enough for policy wording.
Ornith 1.5 35B-A3B (BF16, LoRA rank 16)	Similar active size to Nemotron but built as an image-and-text model, which could later read scanned tables in the manuals. Tests whether that broader architecture costs answer quality.
Muse Glimmer 30B (4-bit, LoRA rank 16)	The model already served locally; 4-bit loading cuts memory. Tests whether the existing service can be adapted instead of replaced.

All 16 runs (four families × four sizes) completed with zero test or control rows. Validation loss improved with more training data for Gemma, Nemotron and Ornith and was flat for Muse after 509 records; from 509 records up, every family began to overfit after one or two epochs. Token-level results are in Appendix D. They cannot rank families because each family uses a different tokenizer.

3. Baseline

Current practice. An analyst searches the PDF and drafts the answer by hand. I measured this on my 30 hand-written lookup questions: 30/30 correct, about 84 seconds per question (42 minutes 6 seconds in total, estimated from session timestamps rather than a stopwatch). These are easy page and code lookups on different questions, so this is a ceiling reference for simple lookups, not a same-split score.

Technical baseline on the same split and metric. Each family's untouched base model answered the same 24 validation questions with the same prompt and decoding as its adapter (Table 1). My PM Plan success criterion uses this baseline: the adapter must beat its own base model by at least 10 percentage points.

4. Evidence

Metric. My committed primary metric is human-judged factual accuracy on answerable questions. For the 192 development outputs (four families × base and adapter × 24 questions), a local AI judge compared each answer with the true source page. An answer counts as fully supported only if it answers the question, is correct, is supported by the page and contains no unsupported claim. The secondary metrics follow the cost asymmetry: an answer with an unsupported claim is the costly error (a confident wrong policy statement), while an abstention is safe but unhelpful. Failed generations stay in the denominator as failures.

Table 1. Same 24 validation questions (9 documents), question-only input, greedy decoding, AI-judged

Model	Condition	Fully supported (95% CI)	Unsupported claims	Abstained	Failed outputs
Nemotron	Untouched base	4/24 (7%–36%)	17/24	3	0
Nemotron	Adapter	4/24 (7%–36%)	19/24	0	0
Gemma	Untouched base	0/24 (0%–14%)	10/24	14	0
Gemma	Adapter	4/24 (7%–36%)	18/23	0	1
Ornith	Untouched base	1/24 (1%–20%)	0/24	23	0
Ornith	Adapter	4/24 (7%–36%)	18/23	0	1
Muse	Untouched base	0/24 (0%–14%)	0/8	8	16
Muse	Adapter	0/24 (0%–14%)	—	—	24

Paired with its own base on the same questions, the Nemotron adapter gained one question and lost one (exact McNemar p = 1.0); Gemma gained 4 (p = 0.125); Ornith gained 4 and lost 1 (p = 0.375). None is reliable at this sample size, and all four 95% intervals at 4/24 overlap. Muse produced almost no scorable outputs: its native output format broke the answer parser (23 failures) and 16 base answers ran out of token budget, so Muse was dropped for lack of evidence rather than for low quality.

Table 2. Week 8 human review of the selected candidate: 15 validation questions, Nemotron base vs. adapter

Measure	Untouched base	Adapter
Correct, supported and no unsupported claim (primary, strict)	2/15 (13%)	4/15 (27%)
Factually correct	3/15	4/15
Unsupported claim present	12/15	11/15
Citation given / citation points to the correct page	9/15 / 0/15	15/15 / 0/15

The adapter improved 2 questions and worsened none (p = 0.5). The review was AI-assisted: the judge's suggestion was visible and I confirmed it on all 180 ratings, so this is not an independent blind score. All 15 questions come from Provider Manual chapters. Tuning: no hyperparameters were searched on these questions. LoRA settings were fixed per family; the only selection was the checkpoint with the lowest validation loss on all 124 validation records, which include these 24 questions. That gives the adapters a small selection advantage in Table 1.

5. Selection and Justification

I selected Nemotron 3.5 Lightning trained on 1,600 records, checkpoint 400 (end of epoch 1; lowest validation loss), to be tested once against its untouched base. I made this choice on October 8.

The score cannot decide. Three adapters tie at 4/24 and their intervals overlap completely. The tie-breaks I declared before scoring did not separate them either: answers with unsupported claims were 19/24 (79%) for Nemotron and 18/23 (78%) for Gemma and Ornith, whose 24th outputs failed; the AI judge's citation-error counts proved unreliable (Section 6); and latency was measured only for Nemotron. The decision therefore rests on data characteristics and business constraints:

- **Cleanest test of what fine-tuning adds.** Nemotron's base model already attempts 21 of 24 questions, so its base-versus-adapter comparison measures added knowledge. The Gemma and Ornith gains coincide with their adapters no longer declining (abstentions 14→0 and 23→0) while answers with unsupported claims rose from 10 to 18 and from 0 to 18. Under my cost asymmetry, trading safe refusals for confident errors is not an improvement.
- **Complete, checkable evidence.** Nemotron is the only family with all 48 outputs scored and no failures, and the only one with a human-reviewed sample (Table 2: 2/15 to 4/15).
- **Deployment fit.** About 3 billion active parameters. On one RTX PRO 6000 Blackwell GPU the adapter's median answer time was 2.9 seconds (90th percentile 3.4 seconds), fast enough for interactive drafting, and the loaded model used 63.2 GB of the card's 96 GB.

Risk, stated plainly: on 24 questions the Nemotron adapter showed no gain over its base, and 19 of its 24 answers contained an unsupported claim. It may fail the +10-point target. If it does, I will report a negative result (Section 7).

6. Interpretability and Leakage Check

For a language model, the equivalent of feature importance is asking which output behaviors drive the scores and whether they make business sense. Three findings stand out.

- **Citations are format, not grounding.** After fine-tuning, Nemotron attached a citation to all 15 reviewed answers (base: 9), yet none of the 30 citations pointed to the correct page. Training on URL-bearing records taught the citation format, which is the effect your feedback warned about. I score citation correctness against the true page and never count citation presence as grounding.
- **Abstention disappeared because the training data has no unanswerable examples.** Every adapter stopped declining (Nemotron 3→0, Gemma 14→0, Ornith 23→0). The models learned what the data showed: every question has an answer. That conflicts with the business need to decline when support is missing, so the final test includes unanswerable questions and gives both models the source passage.
- **The AI judge is reliable for correctness, not for citations.** On 12 Nemotron outputs scored by both the AI judge and the human review, they agreed on correctness, support, unsupported claims and the overall pass (12/12), but on citation validity only 3/12: the judge accepted wrong citations. Citation correctness stays a human-checked metric.

Leakage hunted and addressed

Check	Finding	Action
Documents shared across splits	0 (file-fingerprint check).	Split is grouped by document.
Identical answers across splits	16 groups, 57 records. My October 5–6 rulings: 6 harmless shared values (such as "page 2"), 3 boilerplate, 7 substantive overlaps (18 records), plus 1 substantive near-duplicate question pair (Appendix C).	The 20 affected records (11 train, 4 validation, 5 test) will be corrected in a versioned copy of the split before any reuse. None of the 24 development questions or 15 reviewed questions is in a substantive group, so Tables 1–2 are unaffected.
Control questions vs. training documents	All 30 hand-written controls come from test documents; none from train or validation (Appendix B).	Answers the memorization concern in your feedback.
Exposed historical test	The 191 test records were used for diagnostics.	The final test uses fresh official documents. An October 14 audit found no exposure evidence for 11 of 12 candidate documents and flagged one pharmacy chapter for review.
Source-reading error	One generated record misread a table value (H2011 daily maximum).	The correct value, 4, will go into the new data version.

7. What Would Change My Mind

Before the final test: if, given the source passage, the Nemotron adapter produced more unsupported claims than its base on development questions, or if a human review showed Gemma or Ornith making fewer unsupported claims at equal accuracy, I would switch candidates before locking the test. After the final test: the adapter is not promoted unless it beats

its untouched base by at least 10 points on answerable accuracy, with at least 95% correct citations, fewer than 2% unsupported claims, at least 90% correct abstention, at least 95% correct safety routing, no more than a 3-point general-task regression and interactive latency. If it misses, I will not tune on the test or swap in a rival. The evidence so far (no gain on 24 questions, a small gain on 15 reviewed questions) would then point to the untouched base model with the source passage supplied, which is simpler and needs no fine-tuning.

8. Repository

Course repository (public): github.com/porespellar/INAI6986_project_deliverables

Notebook: notebook/current_evidence_replay_20261018.ipynb

The notebook runs top to bottom offline with zero errors. Starting from the prepared records and the stored per-output judgments, it verifies every input fingerprint, then recomputes the split and leakage checks, the 16 training summaries, Tables 1 and 2, the judge agreement check and the duplicate-group tally. Steps that need the GPU server (record generation, training and model answers) are not re-executed; their stored outputs are fingerprint-checked. A final-test notebook will follow the one-time test.

Appendix A — AI Use Disclosure

I used Hermes Agent with GPT-6 Luna and GPT-6.1 Sol (October 1–16) and Claude Opus 5.5 (October 18) to draft and revise this report and to write the analysis and document-building code. Local Qwen models generated the candidate training records and the advisory development scores. I designed the project; chose the models, metrics and split; wrote the 30 control questions; made the duplicate rulings and the Week 8 review ratings; and selected the candidate. I checked the reported figures against the notebook output and remain responsible for the content.

Appendix B — Hand-Written Control Questions and Their Split

My 30 source-checked lookup questions, preserved verbatim. Every source document is in the test split (matched by full URL and SHA-256 against dataset f0d9324c68e7865f…); none is in train or validation. The three most used pages account for 21 of the 30 questions, so this set measures simple lookups on a few pages.

ID	Question	Expected answer	Source PDF / page	Split
SA-001	What page number in the table of contents lists Rule 560-X-1-.07, Provider Rights and Responsibilities?	Page 3.	9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1	test
SA-002	What page number in the table of contents lists Rule 560-X-28-.02, Summary of State Law Regarding Advanced Directives?	Page 2.	9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1	test
SA-003	What page number in the table of contents lists Rule 560-X-49-.07, Copayment?	Page 2.	9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1	test
SA-004	What page number in the table of contents lists Rule 560-X-50-.03, Recipient Eligibility?	Page 3.	9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1	test
SA-005	What Medicaid provider type is assigned to Licensed Social Workers?	Provider type 11, Crossovers Only.	Apr26_23.pdf p.1	test
SA-006	What provider type is assigned to therapy providers?	Provider type 17, Therapy.	Apr26_37.pdf p.2	test
SA-007	What provider type is assigned to Remote Patient Monitoring providers?	Provider type 08, Remote Patient Monitoring.	Jul26_111.pdf p.1	test
SA-008	What provider type is assigned to physicians?	Provider type 31, Physician.	Jul26_28.pdf p.2	test
SA-009	What provider type is assigned to preventive health education providers?	Provider type 55.	Jul26_30.pdf p.1	test
SA-010	What provider type is assigned to psychology providers?	Provider type 54, Psychologist.	Jul26_34.pdf p.2	test
SA-011	What provider type is assigned to Behavioral Health providers?	Provider type 07, Behavioral Health.	Jul26_34.pdf p.2	test
SA-012	What specialty code is listed for Psychology providers?	Psychology specialty code 112.	Jul26_34.pdf p.2	test
SA-013	What specialty code is listed for a Licensed Professional Counselor / LPC?	Specialty code 070	Jul26_34.pdf p.2	test
SA-014	What provider type is assigned to Nurse Practitioners?	Provider type 09.	Jul26_28.pdf p.2	test
SA-015	What provider type is assigned to Physician Assistants?	Provider type 10.	Jul26_28.pdf p.2	test
SA-016	What specialty code is listed for Occupational Therapy?	Occupational Therapy code 171.	Apr26_37.pdf p.2	test

ID	Question	Expected answer	Source PDF / page	Split
SA-017	What specialty code is listed for Physical Therapy?	Physical Therapy code 170.	Apr26_37.pdf p.2	test
SA-018	What specialty code is listed for Speech Therapy?	Speech Therapy code 173.	Apr26_37.pdf p.2	test
SA-019	What specialty code is listed for Applied Behavior Analysis Therapy.	Applied Behavior Analysis (ABA) Therapy. Code 175.	Apr26_37.pdf p.2	test
SA-020	What provider specialties are assigned to Remote Patient Monitoring providers?	Provider specialties 085 or 086.	Jul26_111.pdf p.1	test
SA-021	What specialty code is listed for an Associate Licensed Counselor / ALC?	Specialty code 071.	Jul26_34.pdf p.2	test
SA-022	What specialty code is listed for a Licensed Marriage and Family Therapist / LMFT?	Specialty code 072.	Jul26_34.pdf p.2	test
SA-023	What specialty code is listed for a Licensed Marriage and Family Therapist Associate / LMFTA?	Specialty code 075.	Jul26_34.pdf p.2	test
SA-024	What specialty code is listed for a Licensed Master Social Worker / LMSW?	Specialty code 073.	Jul26_34.pdf p.2	test
SA-025	What specialty code is listed for a Licensed Independent Clinical Social Worker / LICSW.	Specialty code 074.	Jul26_34.pdf p.2	test
SA-026	What specialty code is listed for a Licensed Psychological Technician / LPT?	Specialty code 076.	Jul26_34.pdf p.2	test
SA-027	What specialty code is listed for Preventive Health Education?	Specialty code 183.	Jul26_30.pdf p.1	test
SA-028	What specialty code is listed for EPSDT in the physician specialty table?	EPSDT code 560.	Jul26_28.pdf p.2	test
SA-029	What specialty code is listed for Family Practice in the physician specialty table?	Family Practice code 316.	Jul26_28.pdf p.2	test
SA-030	What specialty code is listed for Cardiac Surgery in the physician specialty table?	Cardiac Surgery code 312.	Jul26_28.pdf p.2	test

Questions per page: Jul26_34.pdf p.2: 10; Jul26_28.pdf p.2: 6; Apr26_37.pdf p.2: 5; Jul26_111.pdf p.1: 2; Jul26_30.pdf p.1: 2; 9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1: 1; 9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1: 1; 9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1: 1; 9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1: 1; Apr26_23.pdf p.1: 1.

Appendix C — Cross-Split Duplicate Answers and My Rulings

Group	Shared answer (normalized, abbreviated)	Records	Splits	My ruling
G01	page 2	8	test 1 / train 5 / validation 2	Harmless Shared Value
G02	this page intentionally left blank	7	test 1 / train 5 / validation 1	Boilerplate Only
G03	the current procedural terminology cpt and current dental ...	6	train 4 / validation 2	Boilerplate Only
G04	page 3	5	test 1 / train 3 / validation 1	Harmless Shared Value
G05	3 90	4	test 1 / train 3	Substantive Overlap
G06	icd 9 codes should be used for claims submitted with dates...	4	test 1 / train 2 / validation 1	Substantive Overlap
G07	3	3	test 1 / train 2	Harmless Shared Value
G08	4	3	train 2 / validation 1	Harmless Shared Value
G09	chapter 5	3	train 2 / validation 1	Harmless Shared Value
G10	11	2	test 1 / train 1	Harmless Shared Value
G11	334 215 7416	2	test 1 / train 1	Substantive Overlap
G12	a hard copy ub 04 claim form must be submitted	2	train 1 / validation 1	Substantive Overlap
G13	the amount billed shall not exceed the entity s actual acq...	2	test 1 / train 1	Substantive Overlap
G14	the current procedural terminology cpt and current dental ...	2	train 1 / validation 1	Boilerplate Only
G15	the ltc gainwell cover sheet from the web portal must acco...	2	train 1 / validation 1	Substantive Overlap
G16	up to four procedure code modifiers	2	test 1 / train 1	Substantive Overlap
ND01	near-duplicate question pair (train / validation)	2	train 1 / validation 1	Substantive overlap

Harmless shared values are coincidences such as the same page number in different chapters; boilerplate is repeated template text. Substantive overlaps repeat the same policy fact across splits and are the only groups that need a remedy.

Appendix D — Training Runs (Token-Level Diagnostics Only)

Teacher-forced next-token metrics on the fixed 124 validation records. They show how well each run fits its own tokens and are not answer accuracy; values are not comparable across families.

N = 1,600 runs: final step, selected checkpoint and loss / next-token accuracy

Family	Final step	Selected step	Final train loss / accuracy	Selected validation loss / accuracy
Gemma	600	400	0.2435 / 93.05%	0.5750 / 85.88%
Muse	600	200	0.0877 / 97.56%	0.5787 / 85.54%
Nemotron	1200	400	0.2265 / 92.98%	0.5527 / 85.56%
Ornith	1200	400	0.1796 / 94.01%	0.5626 / 85.38%

All 52 epoch records from the 16 runs (a dash means the metric was not recorded at that checkpoint)

Model / N	Epoch / step	Train loss	Train accuracy	Val. loss	Val. accuracy
Gemma 4; N=200	1.00 / 25	1.1405	74.60%	—	—
Gemma 4; N=200	2.00 / 50	0.7575	82.99%	0.7751	82.15%
Gemma 4; N=200	3.00 / 75	0.6893	84.40%	0.7044	83.45%
Gemma 4; N=509	1.00 / 64	0.6592	84.73%	0.6508	84.19%
Gemma 4; N=509	2.00 / 128	0.5430	86.60%	0.6060	84.93%
Gemma 4; N=509	3.00 / 192	0.4433	88.52%	0.6067	85.19%
Gemma 4; N=509	3.13 / 200	—	—	0.6178	85.25%
Gemma 4; N=509	4.00 / 256	0.3688	90.06%	0.6321	85.24%
Gemma 4; N=509	5.00 / 320	0.3373	90.81%	0.6636	85.03%
Gemma 4; N=750	1.00 / 94	0.6182	85.26%	0.6199	84.68%
Gemma 4; N=750	2.00 / 188	0.4791	87.69%	0.5829	85.58%
Gemma 4; N=750	3.00 / 282	0.3361	90.63%	0.6205	85.37%
Gemma 4; N=750	4.00 / 376	0.2192	93.62%	0.6946	85.13%
Gemma 4; N=1,600	1.00 / 200	0.5400	86.69%	0.5751	85.79%
Gemma 4; N=1,600	2.00 / 400	0.3832	89.63%	0.5750	85.88%
Gemma 4; N=1,600	3.00 / 600	0.2435	93.05%	0.6180	85.65%
Muse Glimmer; N=200	1.00 / 25	0.5252	86.63%	0.5853	85.10%
Muse Glimmer; N=200	2.00 / 50	0.3505	90.32%	0.6073	84.98%
Muse Glimmer; N=200	3.00 / 75	0.2376	93.15%	0.6589	84.54%
Muse Glimmer; N=509	1.00 / 64	0.4638	87.86%	0.5727	85.44%
Muse Glimmer; N=509	2.00 / 128	0.2305	93.62%	0.6264	84.86%
Muse Glimmer; N=509	3.00 / 192	0.0856	97.64%	0.7456	84.70%
Muse Glimmer; N=750	1.00 / 94	0.4532	88.04%	0.5725	85.44%
Muse Glimmer; N=750	2.00 / 188	0.2137	93.82%	0.6279	85.32%
Muse Glimmer; N=750	3.00 / 282	0.0846	97.70%	0.7447	84.80%
Muse Glimmer; N=1,600	1.00 / 200	0.4067	89.15%	0.5787	85.54%
Muse Glimmer; N=1,600	2.00 / 400	0.1969	94.23%	0.6054	85.41%
Muse Glimmer; N=1,600	3.00 / 600	0.0877	97.56%	0.7249	84.72%
Nemotron; N=200	1 / 50	0.5403	85.57%	0.5859	84.73%
Nemotron; N=200	2 / 100	0.4232	88.02%	0.5825	84.75%
Nemotron; N=200	3 / 150	0.3895	88.65%	0.5862	84.82%
Nemotron; N=509	1 / 128	0.4874	86.59%	0.5614	85.25%
Nemotron; N=509	2 / 256	0.3613	89.43%	0.5637	85.30%
Nemotron; N=509	3 / 384	0.2429	92.52%	0.6118	84.83%
Nemotron; N=750	1 / 188	0.4774	86.66%	0.5574	85.23%
Nemotron; N=750	2 / 376	0.3577	89.46%	0.5593	85.37%
Nemotron; N=750	3 / 564	0.2363	92.61%	0.6060	85.17%

Model / N	Epoch / step	Train loss	Train accuracy	Val. loss	Val. accuracy
Nemotron; N=1,600	1 / 400	0.4671	87.09%	0.5527	85.56%
Nemotron; N=1,600	2 / 800	0.3509	89.76%	0.5549	85.49%
Nemotron; N=1,600	3 / 1200	0.2265	92.98%	0.6001	85.15%
Ornith; N=200	1 / 50	0.5132	86.06%	0.6011	84.17%
Ornith; N=200	2 / 100	0.3855	88.72%	0.6028	84.62%
Ornith; N=200	3 / 150	0.3454	89.65%	0.6184	84.57%
Ornith; N=509	1 / 128	0.4750	86.99%	0.5822	84.96%
Ornith; N=509	2 / 256	0.3272	90.14%	0.5829	84.85%
Ornith; N=509	3 / 384	0.1945	93.56%	0.6390	84.89%
Ornith; N=750	1 / 188	0.4683	86.92%	0.5809	84.98%
Ornith; N=750	2 / 376	0.3251	90.16%	0.5834	85.17%
Ornith; N=750	3 / 564	0.1857	93.92%	0.6355	84.68%
Ornith; N=1,600	1 / 400	0.4613	87.24%	0.5626	85.38%
Ornith; N=1,600	2 / 800	0.3083	90.62%	0.5797	85.46%
Ornith; N=1,600	3 / 1200	0.1796	94.01%	0.6254	85.21%

Appendix E — Evidence Files

Claim	File in the source-of-record repository
Prepared dataset and split	artifacts/milestone1/2026-09-17/approved_candidates.jsonl (SHA-256 f0d9324c68e7865f…)
Hand-written controls	data/curated/student-authored-control-set-2026-09-22.jsonl (SHA-256 a1deb52304d7bce9…)
Training runs	artifacts/model_comparison/2026-09-25/results.jsonl; architecture-training-gates/clean-runs/architecture-clean-run-matrix.json; run_log.csv (16 runs)
Table 1 outputs and judgments	artifacts/milestone2_completion/development/shortlist_v1/ (predeclared question list, settings and the 192 counted outcomes)
Table 2 human review	artifacts/milestone2_completion/review/batch_b/verification/submitted_revision_000011/
Duplicate rulings	artifacts/milestone2_completion/review/responses/ (Batch A receipt and saved answers)
All derived numbers in this report	artifacts/milestone2_completion/class_package_20261018/development_analysis_20261018.json
Instructor feedback and rubric	docs/engineering/course-instructions/

Appendix F — Response to Instructor Feedback

Your feedback	What I did	What is still open
Controls are page/code lookups, not policy support	Kept the 30 originals unchanged and reported them as a lookup ceiling (Section 3).	The fresh final benchmark adds exception, date and multi-rule questions. Its questions are written by a local AI model and source-checked, not by me; I disclose that change.
No unanswerable questions	Submitted nine extra unsupported-question pairs (eight adopted AI drafts, one edited; labelled as AI-assisted).	The final benchmark includes an unanswerable stratum so abstention can be measured.
20 of 30 from about four pages	Showed the concentration in Appendix B (21 of 30 on three pages).	The final benchmark spreads questions across documents.
Are the control documents in training?	Mapped all 30: every source is a test document; none in train or validation.	None.
Manual 30/30 is a ceiling	Reported it as a ceiling; the same-split baseline is the untouched base model (Table 1).	A timed analyst baseline on harder questions was not run.
Citation presence is format acquisition	Confirmed: the adapter cites on 15 of 15 answers, 0 of 30 citations are correct (Section 6).	None; citation presence is no longer used as evidence.
57 cross-split duplicate records undispositioned	Ruled on all 16 groups and the near-duplicate pair (Section 6, Appendix C).	Apply the remedy in a versioned split before any reuse.
Candidates A and B were one recipe	Trained four genuine families at four sizes (16 runs).	None.
Thirty single-record control pages	Collapsed into one table (Appendix B).	None.